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Försättsblad Prov Original

Kurskod	Provkod	Tentamensdatum
DT081A	T101	2026-01-07
Kursnamn	Datateknik AV, Signal- och bildbehandling	
Provnamn	Tentamen	
Ort	Sundsvall	
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Ämne		

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2026-01-07. Examination in

DT081A – Signal and Image Processing, 6 hp

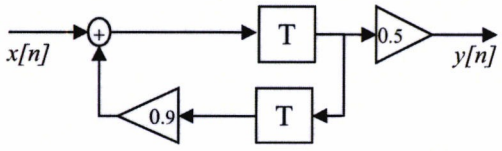
Allowed aids	Dictionary, non-programmable calculator and drawing tools (e.g ruler), Physics Handbook, TeFyMa, and formula collection booklet for mathematics/physics courses at MIUN. A collection of formulas is also presented at the end of this exam.
Miscellaneous	Don't forget to: 1. Submit one, or several pages, per solved assignment and NOT several solved assignments per page. 2. Order the handed in pages with the solved assignments in ascending order. 3. Describe and clearly motivate all solutions, assumptions, programming constructs, etc.. 4. Attach all neatly drawn figures that you might want to refer to in a solution. **Failing to follow the above instructions will render point reduction.**

Good Luck!

1. What do the following signal processing concepts mean? Be clear, concise, and relevant.

a) Convolution (1p)
b) Aliasing (1p)
c) LTI system (1p)
d) Spectrogram (1p)
e) Analogue signal (1p)
f) Reconstruction (1p)

2. A time discrete system is defined as below.



a) Using the block diagram, explain whether the system is FIR or IIR, causal or non-causal, and justify your answer based on its structure. (2p)
b) Derive the system's difference equation from the block diagram. (2p)

3. Given a discrete-time system described by the difference equation

$$y(n+1) + y(n) = x(n) + 0.8y(n-1) - 0.2\mathrm{x(n-1)}$$

a) Identify the type of response that the system produces by the input

$$x(n) = \begin{cases} 1, & n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$ (1p)

b) Calculate the first 3 samples of the output signal $y(0), y(1), y(2)$ by the input

$$x[n] = \begin{cases} 0.8^n \cdot \sin[n+2], & n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$ (2p)

c) Another system has the frequency function

$$H(e^{j\Omega}) = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})} = \frac{1}{1-e^{-j\Omega}-0.5e^{-2j\Omega}} = \frac{e^{2j\Omega}}{e^{2j\Omega}-e^{j\Omega}-0.5}.$$

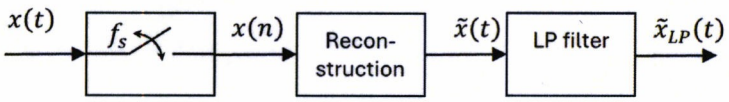
Sketch the amplitude magnification $|H(e^{j\Omega})|$ for frequency $0 \leq \Omega \leq 2\pi$. (3p)

4. The signal $x(t) = 3\cos(50\pi t) + 4\sin(20\pi t - 0.1)$ is sampled and reconstructed according to the figure below.

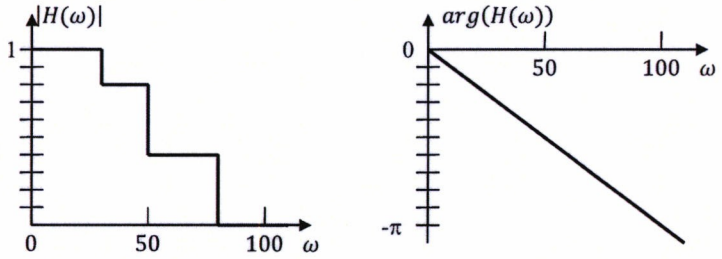
a) What is the lowest sampling frequency f_s [Hz] to enable a full reconstruction of the signal with the same frequency contents? (1p)

b) If the signal is sampled with $f_s = 30\ Hz$, reconstructed and filtered with an ideal LP-filter with cut-off frequency $f_c = 30\ Hz$, what frequencies does the output signal $\tilde{x}_{LP}(t)$ then contain. [Hint: there is aliasing] (2p)

c) Draw the amplitude spectrum for the signal $x(t)$ and sketch (in the same diagram) the frequency response for a real (non-ideal) antialiasing filter that allow full reconstruction of the signal. (1p)

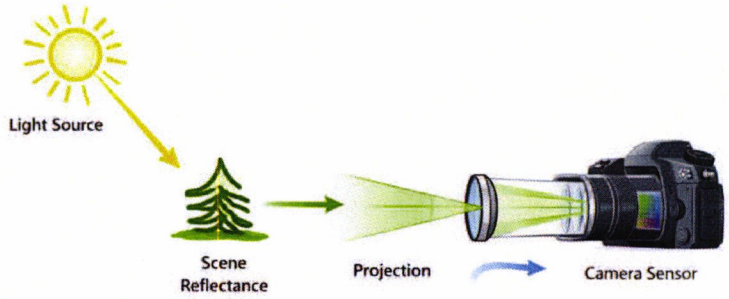


5. A linear, time-invariant (LTI) system is defined by the amplitude and phase spectra in the figure below. The system input signal is
$x(t) = 3\cos\left(15\pi t - \frac{\pi}{10}\right) + 1 + 2\sin(20\pi t + 0.1)$.



a) Decide the output signal $y(t)$. (2p)
b) What kind of filter is the system? (1p)

6. The figure below provides a summary of the image capture process from scene to stored pixel values.



a) Starting from a continuous light field emitted by a source, describe the sequence of transformations that lead to discrete RGB pixel values stored in memory. (3p)
b) After exposure, a digital camera applies a sequence of signal processing operations before pixel values are stored or compressed. Name at least three such operations performed inside the camera and briefly explain the purpose of each operation. (3p)

7. Linear image operators can be analyzed by studying how they act on elementary image components and how these responses combine to form the output image.

a) Starting from the definition of a linear image operator acting on a point source $O\{\delta(x-\alpha, y-\beta)\} = h(x, y, \alpha, \beta)$
explain why a general image can be interpreted as a weighted sum of point sources, and how this leads to the operator expression for the output image $g(\alpha, \beta)$. (3p)

b) Assume the PSF is shift-invariant. Derive the simplified expression for the output image $g(\alpha, \beta)$ and explain why this can be implemented using convolution. (3p)

8. Contrast enhancement aims to improve the visual appearance of an image by modifying its intensity distribution. The figure below shows an original image (a) and an enhanced version (b). Plots (c–f) show different representations related to these two images. Describe what information each of the plots (c–f) conveys and explain, using equations where relevant, how they are involved in producing the enhanced image (b). (5p)

